Lucas Greenwood

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Education

University of Florida Gainesville, FL
Computer Engineering (BS) & Minor in Mathematics **GPA: 3.94/4.00** *Aug. 2024 – May. 2028*

Florida Atlantic University Boca Raton, FL
Certificate in Financial Technology *Jan. 2023 – May. 2024*

Technical Skills

Languages: CPP, Bash, Python, VHDL, AVR, RISC-V, C, Wolfram Language, TypeScript, JavaScript
Tools: Arch Linux, Quartus, Microchip Studio, React, TailwindCSS, Vite, GitHub

Projects

Guru | *SSH, Arch Linux (ARM), Web Scraping* Sept 2025 - Present

- Deployed a Linux OS on a Raspberry Pi, configuring **cron** jobs to fetch content from a remote **SSH** server.
- Implemented a **web-scraping** pipeline that parsed **9** webpages and enqueued **66** meditation mp3s for playback.
- Developed an embedded system driving a speaker through **GPIO** pins, enabling playback via a **touch sensor**.

MeshMatcher | *CPP, Open3D, JupyterLab* Aug 2025 - Present

- Performed exploratory data analysis on the ModelNet **10-class** dataset with **Open3D** and **Jupyter Notebook**.
- Wrote **CPP scripts** to iterate over a directory and process **5000+ .OFF** files and **9507** vertices on average.
- Built an Octree data structure in **CPP** and analyzed the complexity of the class methods.

Deducing Grid Styles Using Neural Networks | *Wolfram Language (WL), Ubuntu* Jun 2024 – Jul 2024

- Built a Vision and Text **Transformer Model** and an **LSTM/CNN** Model from scratch in **WL**.
- Wrote scripts to generate **10,000+** data pairs and train network on server with validation error of **4.58%**.
- Developed vocabularies and utilized character-by-character and **sub-word-token text generation** methods.

Experience

Bike Mechanic Jun 2025 – Present
SG Bike Repair *Gainesville, FL*

- Communicated technical concepts clearly and tailored demonstrations to educate customers on maintenance.
- Familiarized with standardized bike parts, tolerances, and fittings across **30+** models of bike using documentation to support repair and assembly.

Student Researcher Jun 2024 – Jul 2024
Wolfram Summer School *Waltham, MA*

- Animated demonstrative visuals in **Wolfram Language**, presented results orally, and discussed design process of *Deducing Grid Styles Using Neural Networks* in a poster session.
- Compiled feedback from oral presentation and elaborated on technicalities of project design in a final report.

Involvements

Student May 2025 – Present
Semiconductor Career Readiness Organization : Summer Device Processing Lab *Gainesville, FL*

- Gained hands-on nanofabrication experience developing **MOSCAPs** from doped silicon wafers using thermal oxidation, photolithography, wet etching, metal deposition, and annealing.
- Demonstrated competency with cleanroom processes and **device physics** by refining feature sizes through iterative tuning of spin coating rpm, exposure time, and development time.
- Measured and validated **MOSCAP** performance through capacitance–voltage (C–V) characterization with a microprobe **CVU**.

Software Engineer Sep 2024 – Present
Dream Team Engineering : DRIFT Team *Gainesville, FL*

- Developed a dynamic website driven by **GeoJSON**-based APIs and interfaced with backend servers running **SEIRD** disease-prediction algorithms.
- Proposed circuitry modifications and prepared CAD models in **SolidWorks** to prototype an infant leg prosthetic.
- Prepared written proposals for modifications, sketch documentation for CAD parts, and code documentation to streamline communication across team divisions.

Professional Affairs Ambassador Sep 2024 – May 2025
Freshman Leadership Engineering Group *Gainesville, FL*

- Organized a professional development Q&A panel with J&J Vision; drawing **25+** attendees.
- Prepared application banks for an Apply-A-Thon event with **35+** attendees.
- Researched effective cover letter properties and applied findings to a writing workshop for **15+** attendees.